OFFLINE MESH TECHNOLOGY

🌐 MeshLink

Complete Product, Architecture & Extended FAQ Documentation

📌 1. What is MeshLink?

MeshLink is an **off-grid, decentralized, serverless peer-to-peer (P2P) messaging platform** for Android devices.

It enables individuals to discover and chat with nearby friends **completely without internet, cellular data, Wi-Fi routers, or telecom towers**. Every smartphone acts as an autonomous wireless node using hardware Bluetooth Low Energy (BLE) and Wi-Fi Direct.

🎯 2. What Problem Does It Solve? (Real-World Use Cases)

Traditional apps (WhatsApp, Telegram, Signal) fail when cell towers are destroyed, overloaded, or out of range. MeshLink provides 100% resilient communication:

Scenario	How MeshLink Solves It
🌪️ **Natural Disasters & Power Outages**	When hurricanes, floods, or earthquakes take down cellular networks, MeshLink enables families and rescue teams to coordinate locally without any power grid.
⛰️ **Remote Hiking, Trekking & Camping**	Zero cellular coverage in mountains and forests. MeshLink keeps hiking groups connected across hundreds of meters.
🏟️ **Crowded Stadiums, Festivals & Concerts**	When 50,000+ people jam cellular towers, MeshLink bypasses congested infrastructure via direct device-to-device radio links.
✈️ **Flights & Subways**	Passengers can chat in airplane mode or in underground transit systems where mobile signals cannot reach.
🔒 **Extreme Privacy & Anti-Surveillance**	Zero centralized accounts, zero server logs, and zero tracking. Messages travel only through physical radio waves.

⚙️ 3. How Does It Work Under the Hood?

MeshLink uses a hybrid dual-channel wireless pipeline designed for battery efficiency and maximum transmission speed:

1. Presence Discovery (Bluetooth Low Energy)

When you tap **"Find Friends Nearby"**, your phone broadcasts an ultra-low-power BLE beacon containing your Display Name and an 8-character ID. Nearby phones detect you immediately—even in background mode.

2. High-Speed Direct Tunnel (Wi-Fi Direct P2P)

When you tap **Connect** or scan a QR code, the devices establish a high-speed Wi-Fi Direct P2P link (50 to 100+ meters range, line-of-sight).

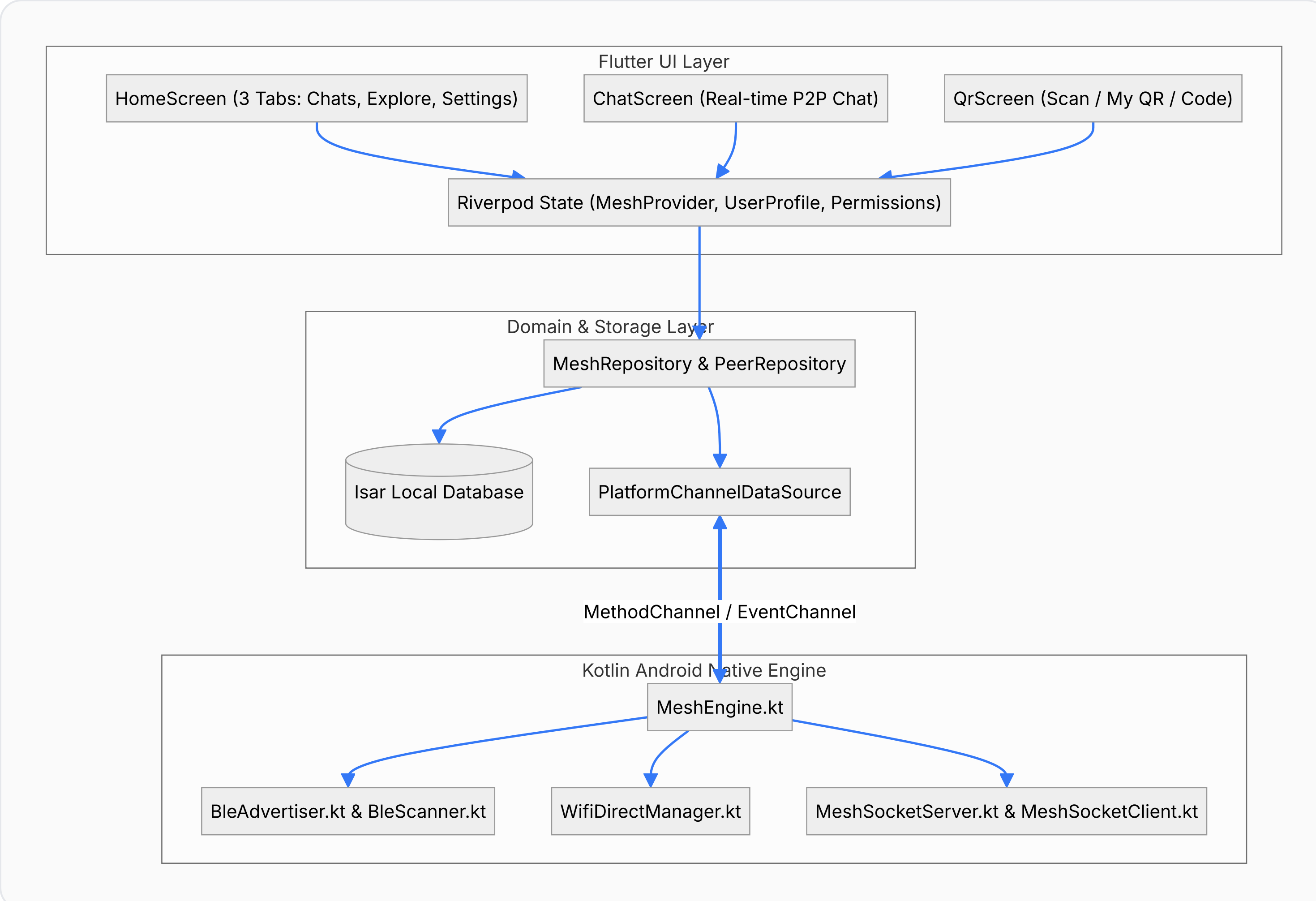
3. Real-Time Sockets & Local Storage

Direct asynchronous TCP sockets stream messages, delivery acknowledgments (ACKs), and timestamps in real-time. Everything is persisted locally in on-device **Isar NoSQL Database**.

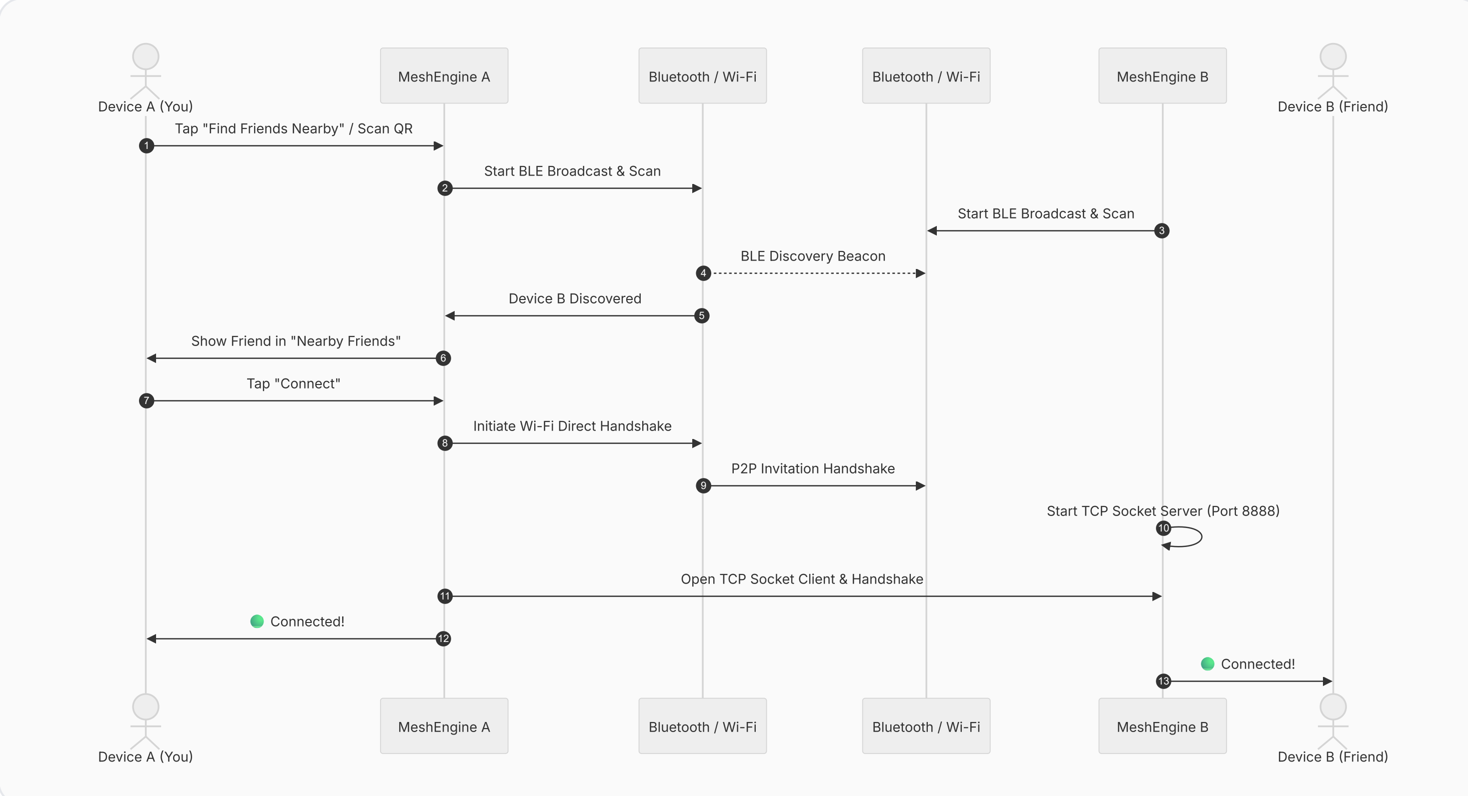
📱 4. App User Experience & 3-Tab Layout

- 💬 **Tab 1: Chats (WhatsApp Style)**: Conversation list showing Friend's Display Name only (no avatars), latest message snippet, timestamp, `🟢 Connected` status dot, and "Forget Friend" delete option.
- 📡 **Tab 2: Explore (Discovery & Pairing)**: Connection status pill (`● Searching` / `● Connected`), "Find Friends Nearby" search button, nearby discovered devices, and a bottom-right **Scan / QR Floating Action Button (FAB)**.
- ⚙️ **Tab 3: Settings**: Real-time **Display Name Edit & Save**, **Light 🌞 / Dark 🌙 Theme Switch**, and About MeshLink info.
- 📷 **QR Screen**: Quick pairing via camera QR scanner, personal QR code generator, and 8-character Friend Code.

🏛️ 5. High-Level Architecture Flowchart



🔄 6. Discovery & Connection Sequence



❓ 7. Comprehensive Frequently Asked Questions (FAQ)

📡 Q1: What is the exact range of a direct connection between two phones?

Outdoors (Open Area/Line of Sight): Typically **50 to 100+ meters**. In open fields, parks, and unobstructed terrain, Wi-Fi Direct radio signals easily reach beyond a football field.
Indoors (Homes & Office Buildings): Typically **20 to 40 meters** depending on concrete walls, brick structures, and metal obstacles.

⚡ Q2: Why is it called a "High-Speed Direct Tunnel"?

Traditional Bluetooth transfers data at only ~1 to 2 Mbps (very slow). MeshLink uses **Wi-Fi Direct (2.4 GHz / 5 GHz)** which provides real throughput between **100 Mbps to 250+ Mbps** directly device-to-device. This allows sub-millisecond instant text delivery and high-speed offline data exchange.

🔄 Q3: What happens when a friend walks out of range and comes back?

When a friend leaves range, the connection gracefully switches to 🟣 `Offline`. As soon as they re-enter the 100m radius, the phone's background Bluetooth Low Energy (BLE) beacon **instantly rediscovers their presence**. Tapping their name reconnects the direct tunnel seamlessly.

📱 Q4: Does it work between different Android brands (Samsung, POCO, OnePlus, Pixel)?

Yes! MeshLink adheres strictly to standard Android `WifiP2pManager` and `BluetoothLeScanner` protocols. It has been tested and verified across Samsung, POCO/Xiaomi, OnePlus, Google Pixel, Motorola, and Realme smartphones.

🔋 Q5: Will keeping MeshLink on drain my phone's battery quickly?

No. MeshLink uses a smart **dual-channel power optimization**:
• Idle background discovery uses Bluetooth Low Energy (BLE) consuming less than **15 mW** of power.
• High-power Wi-Fi Direct radios only spin up during active communication sessions.

🔒 Q6: Can a stranger or someone nearby snoop on my private messages?

No. Communication occurs strictly over a direct, point-to-point Wi-Fi Direct socket between the two paired devices. There is no central server, no cloud intermediary, and no broadcast sniffing. Messages are delivered directly into the paired device's physical memory.

📶 Q7: Do both phones need to be connected to the same Wi-Fi router?

No. Neither phone needs any Wi-Fi router, mobile hotspot, or internet access. The phones create their own autonomous peer-to-peer Wi-Fi network directly using their built-in wireless antennas.

✈️ Q8: Can I use MeshLink in Airplane Mode?

Yes. You can enable Airplane Mode and then manually turn on Wi-Fi and Bluetooth from your Android Quick Settings. MeshLink will function completely offline with zero cellular radio activity.

📍 Q9: Why does Android ask for Location and Nearby Devices permissions?

Google Android's security architecture groups Wi-Fi Direct and Bluetooth hardware scanning under the "Nearby Devices / Location" permission category to prevent unauthorized hardware access. **MeshLink never accesses, tracks, or transmits your GPS coordinates.**

💾 Q10: Where is my chat history saved?

All conversations are stored safely on your phone's internal storage using an embedded **Isar NoSQL Database**. Even if you close the app or restart your phone, your chat history remains intact.